Structural Priors and Architectural Syntax: TBD

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Abstract

Many deep learning architectures generalize on tasks by respecting certain mathematical structure. In the most general sense, a prior in deep learning specifies the nature of this structure, and which transformations of inputs and outputs a model must respect. We propose a theory of such priors, studied as a mathematical objects $(\mathcal{S}, X, Y)$, where $\mathcal{S}$ specifies the structure of the prior, and where $X, Y : \mathcal{S} \to \mathcal{C}$ are functors specifying how the input and output spaces of a model $f : X(x) \to Y(y)$ should transform under the prior. The set of admissible models that "respect the prior" are natural transformations $X \Rightarrow Y$.

1 Introduction

1.1 The history of priors in deep learning

Many neural network architectures have relatively recently been identified with specific mathematical structures, in the sense that they satisfy a compatibility between two ways of processing data: first transform the input data and then predict, or first predict and then transform the output.

In Geometric Deep Learning (GDL) [Bronstein et al., 2021], the mathematical structure is that of a group G, and transformations are various group actions $\rho(g)$ on the input and output spaces. This yields conditions like group equivariance, where a model $f_p : X \to Y$ is said to be G-equivariant if $f \circ \rho_X(g) = \rho_Y(g) \circ f$. If the output transformation ρ_Y is the identity transformation, then one retrieves G-invariance. GDL has so far been highly successful reducing the burden of the curse of dimensionality, as these constraints result in weight-tying schemes, which in turn reduces the number of trainable parameters. Moreover, GDL has given a certain epistemic and scientific value, offering explanations of *why* certain architectures work better than other ones for specific tasks. The classical example of this is that convolution neural networks are rotation-invariant, so rotating an image of a dog and then making a prediction about, say what race the dog is, yields the same output as not rotating the image at all. So why does CNNs work so well for image-classification? One plausible explanation is that because they are rotation-invariant! These are examples of what we will continuously and generally refer to as *priors*, in this case, *geometric* priors.

Compatibility with logical theories is another prior [Nye, 2025]. The mathematical structure can be a Lawvere theory, generated by a set of logical connectives and axioms, such as the law of excluded middle, De Morgan's laws, conjunction associativity and commutativity, or modens tollens. This can help us reduce hallucinations and logical fallacies. We refer to these as *logical priors*.

In Categorical Deep Learning (CDL) [Gavranović et al., 2024],

In Maruyama [2025], many forms of are studied, such as X, and they all follow the same pattern, even when the relevant transformations are not invertible [Gavranović et al., 2024].

In this paper we embrace these patterns and package them as a mathematical triple, consisting of a specification (a group, Lawvere theory, or a monoid), and we study this abstraction and its mathematical properties in its own right. We then investigate the connections between priors and architectural syntax. The structure to preserve itself cannot generally be specified by a single mathematical object, like an abstract symmetry group or general algebraic theories. One must also say how that structure acts on inputs and outputs. This motivates the triple

$$\Pi = (\mathcal{S}, X, Y), \quad X, Y : \mathcal{S} \to \mathcal{C}.$$

The category $\mathcal{S}$ describes types and permitted structural transformations. The functors X and Y interpret them on the input and output sides. A natural transformation between these interpretations is a structure-preserving predictor.

1.2 When category theory becomes useful

1.3 Contribution and related work

Equivariance as naturality appears in [de Haan et al., 2020], where Natural Graph Networks (NGNs) express equivariant kernels through natural transformations. CDL relates equivariance, monad-algebra homomorphisms, and parameterized implementations. The most closely related work is Categorical Equivariant Neural Networks (CENNs) [Maruyama, 2025]. In CENNs, equivariance is through structures extending beyond global group actions. He develops convolutional and nonlinear layers and claims a conditional density theorem for continuous natural transformations.

Consequently, the main contribution is a calculus of priors and their realizations, not the bare observation that natural transformations express structure preservation. We adopt the established naturality-based formulation of structural compatibility and study structural specifications together with their input and output representations as objects of comparison. Our focus is on morphisms between these data, their induced transport of admissible models, and their extension to parameterized realizations and architectural semantics.

The purpose throughout this paper is, in a way, to formalize and articulate the pattern consistent across priors in deep learning and our different ways of expressing them. We want to make relations between priors precise. By doing so, we construct tools to study questions such as: "What does it mean for a prior to relate to another prior, and does such a relation induce a corresponding change in architectural syntax?".

1.4 Notation and conventions

Functors use capital letters; natural transformations use tiny letters; if categories are not represented in bold text, they are represented using calligraphic, capital letters; a prior subscripted with an index "i" is assumed to denote $\Pi_i = (\mathcal{S}_i, X_i, Y_i)$ by default;

Proofs that don't add to the narrative of the construction of the theory and are routine will be included in appendix A.

2 An elementary theory of priors

2.1 The abstract definition of a prior

Definition 2.1 (Structural priors)**.** Fix a locally small category $\mathcal{C}$. Then, a *structural prior* or simply *prior* is a triple $\Pi = (\mathcal{S}, X, Y)$ where $\mathcal{S}$ is a small category, called the *prior specification*, $X, Y : \mathcal{S} \to \mathcal{C}$ are functors called *input* and *output functors*, respectively.

This definition is incredibly general, but it is the minimal starting point for priors.

Remark 2.2. No closedness, topology, differentiability, or measure is required for this definition. The morphisms of $\mathcal{C}$ determine what kinds of component maps are admitted. In Set they are functions; in Vect they are linear maps; in a category of spaces with continuous maps they are continuous maps. Cartesian closedness does not itself mean "continuous."

Definition 2.3 (admissible models)**.** Let Π be a prior[1]. Its *admissible models*, denoted $\mathcal{H}_\Pi$, is the set of natural transformations between the priors input and output functor $\mathcal{H}_\Pi := \mathrm{Nat}(X, Y)$. We say that any $h \in \mathcal{H}_\Pi$ *respects the prior* or *respects* Π. We refer to X and Y together as the *prior representations*.

That is, any admissible model $f \in \mathcal{H}_\Pi$ consists of components $f_s : X(s) \to Y(s)$ such that for any $u : s \to r$, the following diagram commutes for every object $s \in \mathrm{Ob}(\mathcal{S})$:

$$\begin{array}{ccc} X(s) & \xrightarrow{X(u)} & X(r) \\ {\scriptstyle h_s}\downarrow & & \downarrow{\scriptstyle h_r} \\ Y(s) & \xrightarrow[Y(u)]{} & Y(r) \end{array}$$

If we think of a model or a component of a model as a morphism in $\mathcal{C}$, then we say it's admissible if it extends to the imagine of the evaluation

$$\mathrm{ev}_s : \mathcal{H}_\Pi \longrightarrow \mathcal{C}(X(s), Y(s)), \quad h \longmapsto h_s,$$

at some distinguished type $s \in \mathrm{Ob}(\mathcal{S})$. This evaluation need not to be injective. For finite-product-preserving algebras of a (one-sorted) Lawvere theory, evaluation at the distinguished object determines the entire transformation.

[1] Not to be confused with probability distributions on parameters or functions.

In many examples we will see that we often choose the actions of X and Y on objects such that they line up with the input and output types of our architectures in $\mathcal{C}(A, B)$. We use two different functors because it is not often the case that we want to study priors that act the same way on inputs as outputs (see !invariance-vs-equivariance), not to mention having neural networks with the same input and output types.

For finite-product-preserving models of a one-sorted Lawvere theory, evaluation at the distinguished object determines the entire transformation (see !lawvere-theory-example).

The strongest feature of the definition is that the admissibility of a model depends on both representations, not just on an abstract specification. This specific distinction matters a lot empirically (see) . The objects of $\mathcal{S}$ can index sizes, resolutions, contexts, or data types. Accordingly, one should not confuse the term "model", which is any $h \in \mathcal{H}_\Pi$, i.e., a natural transformation, with that of a "neural network", which indeed is just a single predictor. If the neural network respects Π, then it is *included* in $\mathcal{H}_\Pi$. In this context we think of a model as a family of predictors. For a one-object specification there is just one component (See).

The main limitation is that admissibility alone does not determine how a learner selects among admissible models.

Remark 2.4 (On the smallness of $\mathcal{S}$). One limitation is that we can not describe priors whose specification is simply **Set**.

2.2 Prior morphisms

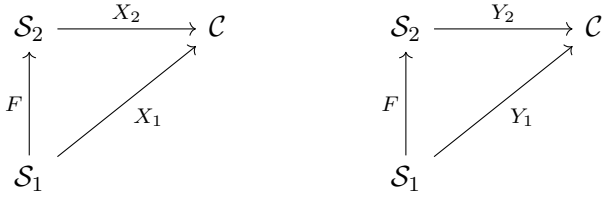
Definition 2.5 (Morphism between priors)**.** For priors $\Pi_1 = (\mathcal{S}_1, X_1, Y_1)$ and $\Pi_2 = (\mathcal{S}_2, X_2, Y_2)$, a *morphism* $\Pi_1 \to \Pi_2$ is a triple $(F, a, b) : \Pi_1 \to \Pi_2$, consisting of a functor $F : \mathcal{S}_1 \to \mathcal{S}_2$ and natural transformations $a : X_1 \Rightarrow X_2 F$ and $b : Y_2 F \Rightarrow Y_1$.

Assuming $\mathcal{H}_{\Pi_2} \neq \emptyset$, there exists a natural transformation $h : X_2 \to Y_2$, and furthermore, if there exists a morphism $\Pi_1 \to \Pi_2$, we have that

$$X_1 \overset{a}{\Rightarrow} X_2 F \overset{hF}{\Rightarrow} Y_2 F \overset{b}{\Rightarrow} Y_1.$$

This means we have an induced natural transformation $h' : X_1 \Rightarrow Y_1$. Note the contravariance here; if $\Pi_1 \to \Pi_2$ exists we can construct $X_1 \Rightarrow Y_1$ via the data of the prior morphism. This gives a semantic result on how prior morphisms translate to functions between admissible models (See Theorem 2.7).

In some situations (**??**), the natural transformations a and b in a prior morphism $(F, a, b) : (\mathcal{S}_1, X_1, Y_1) \to (\mathcal{S}_2, X_2, Y_2)$ reduces to strict equality, in which case the morphism is fully determined by the functor $F : \mathcal{S}_1 \to \mathcal{S}_2$, making the following diagrams strictly commute:



This stricter form of a prior morphism is given by $(F, \mathrm{id}_{X_1}, \mathrm{id}_{Y_1})$. These strict morphisms express exact reindexing of interpretations. They are useful for forgetting operations, restricting

symmetries, and comparing presentations. For instance, in Section 3.1, there are strict morphisms between geometric priors where the specification is expressed using a group action Lawvere theory and a one-object category whose morphisms are isomorphisms. This is because all we need is a change of specification. The "weaker" morphisms add input preprocessing and output postprocessing components. This is seen in Section 3.2, where a morphism from an invariance prior to an equivariance prior is constructed via a specific, invariant output readout.

2.3 A category of priors

Definition 2.6 (A category of priors)**.** The *category of priors* over a locally small category $\mathcal{C}$, denoted $\mathbf{Prior}(\mathcal{C})$ consists of:

- **Objects:** Structural priors over $\mathcal{C}$ (Definition 2.1).
- **Morphisms:** Prior morphisms (Definition 2.5).
- **Composition:** Given two prior morphisms $m := (F, a, b) : \Pi_1 \to \Pi_2$ and $n := (G, c, d) : \Pi_2 \to \Pi_3$, define their *composition* as:

$$n \circ m := (GF, (cF) \circ a, b \circ (dF)).$$

- **Identity:** For a prior $\Pi = (\mathcal{S}, X, Y)$, the *identity morphism* is given by $1_\Pi := (\mathrm{id}_\mathcal{S}, \mathrm{id}_X, \mathrm{id}_Y)$.

For a proof of associativity, unity, and of $\mathbf{Prior}(\mathcal{C})$ being locally small, see Section A.

Note that $\mathbf{Prior}(\mathcal{C})$ is locally small because between two fixed small $\mathcal{S}_1$ and $\mathcal{S}_2$ there is a set of functors F. For each such F, both sets of natural transformations containing a and b are sets by local smallness of $\mathcal{C}$. Their products and the union over f form a set of prior morphisms.

Theorem 2.7 (Contravariance in admissible models)**.** For a morphism of priors $(F, a, b) : \Pi_1 \to \Pi_2$, define $T_m : \mathcal{H}_{\Pi_2} \to \mathcal{H}_{\Pi_1}$ by:

$$T_m(h) = b \circ (hF) \circ a.$$

This mapping $m \mapsto T_m$ together with $\Pi \mapsto \mathcal{H}_\Pi$ defines a functor

$$\mathcal{H} : \mathbf{Prior}(\mathcal{C})^{\mathrm{op}} \longrightarrow \mathbf{Set}.$$

Proof. We first show well-definedness. Consider a $s \in \mathcal{S}_1$ and a $u \in \mathcal{S}_1(s, r)$:

$$(T_m h)_s = b_s \circ h_{F(s)} \circ a_s : X_1(s) \to Y_1(s).$$

Let $u \in \mathcal{S}_1(s, t)$. The natural transformation $T_m h$ is indeed natural:

$$\begin{aligned}
Y_1(u)(T_m h)_s &= Y_1(u)(b_s h_{F(s)} a_s), && \text{(by definition)} \\
&= b_t Y_2(F(u)) h_{F(s)} a_s, && \text{(by naturality of } b\text{)} \\
&= b_t h_{F(t)} X_2(F(u)) a_s, && \text{(by naturality of } h\text{)} \\
&= b_t h_{F(t)} a_t X_1(u), && \text{(by naturality of } a\text{)} \\
&= (T_m h)_t X_1(u).
\end{aligned}$$

Hence, $T_m h \in \mathrm{Nat}(X_1, Y_1)$. Now we check the functoriality axioms. Identity follows since:

$$T_{1_\Pi}(h) = \mathrm{id}_Y \circ (h\mathrm{id}_\mathcal{S}) \circ \mathrm{id}_X = h.$$

Next, let $n = (G, c, d) : \Pi_2 \to \Pi_3$ and $\zeta : X_2 \Rightarrow Y_2$. Then:

$$\begin{aligned} T_{nom}(\zeta) &= b \circ (dF) \circ (\zeta(GF)) \circ (cF) \circ a \\ &= b \circ ((d \circ (\zeta G) \circ c)F) \circ a \\ &= T_m(T_n(\zeta)). \end{aligned}$$

□

Remark 2.8. Note that since **Prior** is locally small, we can apply the Yoneda lemma[2], in which case that

$$\mathrm{Nat}(\mathbf{Prior}_\mathcal{C}(-, \Pi), \mathcal{H}) \cong \mathcal{H}(\Pi).$$

By interpreting this mathematically valid result

By the Yoneda lemma,

2.4 Notions of equivalent priors

Proposition 2.9 (Isomorphisms in the ordinary category)**.** A morphism $(F, a, b) : \Pi_1 \to \Pi_2$ is an isomorphism in $\mathbf{Prior}(\mathcal{C})$ if and only if F is an isomorphism of categories, and a and b are natural isomorphisms.

Proof. See Section A □

Requiring an actual category isomorphism between specifications is often too strict for practical purposes. A standard, and more flexible, choice is the following to loosen the requirement to an equivalence of categories.

Definition 2.10 (Presentation equivalence)**.** A morphism $(F, a, b) : \Pi_1 \to \Pi_2$ is a *presentation equivalence* if F is an equivalence of categories, and a and b are natural isomorphisms.

This gives a useful and more practical notion of what it means for two priors to be "the same" (See Sections 3.1 and 3.3).

Proposition 2.11. A presentation equivalence $m : \Pi_1 \to \Pi_2$ induces a bijection of admissible models:

$$T_m : \mathcal{H}_{\Pi_2} \xrightarrow{\cong} \mathcal{H}_{\Pi_1}.$$

Proof. Precomposition with an equivalence F is an equivalence of functor categories. In particular, it is fully faithful, so

$$\mathrm{Nat}(X_2, Y_2) \longrightarrow \mathrm{Nat}(X_2F, Y_2F), \quad h \longmapsto hF$$

is a bijection. Precomposition by the natural isomorphism a and postcomposition by the natural isomorphism b give a second bijection from this last set to $\mathrm{Nat}(X_1, Y_1)$. Their composite is T_m. □

[2]For a good reference, see Riehl [2017].

2.5 Prior realizations and architectural semantics

Assume now that our base category $\mathcal{C}$ is Cartesian. We have so far defined what it means for a model to respect a prior (Definition 2.3), and we have constructed a contravariant functor $\mathcal{H}$ sending priors to their respective set of admissible models (Theorem 2.7). However, neural networks are not arbtirary maps, nor is it intersting to study admissible maps by itself. The models in deep learning are fundamentally *parameterized maps*. Neural architectures are compositions of parameterized and non-parameterized maps, and so we desire a mathematical object allowing us to study parameterized families of admissible models. We will call these objects "realizations".

Remark 2.12 (On Cartesian structure in deep learning). In deep learning, having $\mathcal{C}$ be Cartesian is empirically backed up by the fact that our classical computers and machine learning models are fundamentally deterministic, so copying and deleting data or information is not a problem. However, although Cartesian products make copying, deletion, branching, and reuse of data natural, but they do not by themselves encode information about tensor axes, dimensionality, contractions, reshaping, or broadcasting, as highlighted in Abbott [2024]. Nonetheless, assuming Cartesianess is, in the field of deep learning, a valid one to make.

Definition 2.13 (Prior realizations)**.** For a parameter space $P \in \mathcal{C}$ and a prior $\Pi = (\mathcal{S}, X, Y)$, the *realization of* Π *using* P is a set of parameterized maps that respect the prior, denoted $\mathcal{R}_\Pi(P)$, defined as:

$$\mathcal{R}_\Pi(P) := \mathrm{Nat}(\Delta P \times X, Y),$$

where $\Delta P : \mathcal{S} \to \mathcal{C}$ is the constant or diagonal functor on P[3].

The functor $\Delta P \times X$ itself is defined by $s \mapsto P \times X(s)$, and $u \mapsto \mathrm{id}_P \times X(u)$, for $u \in \mathcal{S}(s,t)$. An element $\Phi \in \mathcal{R}_\Pi(P)$ consists of parameterized maps $\Phi_s : P \times X(s) \to Y(s)$ such that, for every $u : s \to t$ in $\mathcal{S}$, they respect the prior Π, meaning that the following diagram commutes:

$$\begin{array}{ccc} P \times X(s) & \xrightarrow{\mathrm{id}_P \times X(u)} & P \times X(t) \\ \Big\downarrow{\scriptstyle \Phi_s} & & \Big\downarrow{\scriptstyle \Phi_t} \\ Y(s) & \xrightarrow[Y(u)]{} & Y(t) \end{array}$$

In $\mathcal{C} = \mathbf{Set}$, this idea becomes more transparent:

$$\mathbf{Set}(P, \mathcal{H}_\Pi) \cong \mathrm{Nat}(\Delta P \times X, Y).$$

So specifying a realization in $\mathcal{R}_\Pi(P)$ is equivalent to giving a morphism $r : P \to \mathcal{H}_\Pi$. Thus, we have the interpretation that $\mathcal{H}_\Pi$ is the set of morphisms allowed by the prior, and a realization tells us which part of that set a parameterized model actually implements, and how it is parameterized. For example, suppose our prior Π describes translation-equivariant linear maps, then $\mathcal{H}_\Pi$ would be the set of all translation-equivariant linear maps. But a CNN layer with parameters $p \in \mathbb{R}^k$ gives a parameterization map $r : \mathbb{R}^k \to \mathcal{H}_\Pi$. The prior specifies what is structurally allowed, while the realization specifies how allowed maps are implemented using parameters.

[3]This functor is defined by $s \mapsto P$ for all $s \in \mathcal{S}$, and $u \mapsto \mathrm{id}_P$ for all morphisms $u : s \to t$ in $\mathcal{S}$.

A natural question is: "what happens to realizations of a prior $\mathcal{R}_\Pi(P)$ under a reparameterization $r : P \to Q$?" Given such reparameterization r, we can study the induced reparameterized components of our realization $\mathcal{R}_\Pi(P)$, which we will call the "reparameterized family".

Definition 2.14 (Reparameterized families)**.** Let $\Phi \in \mathcal{R}_\Pi(P)$ be a realized family under a parameter space $P \in \mathcal{C}$, and let $r : P \to Q$ be a 2-morphism (i.e., reparameterization) in $\mathbf{Para}(\mathcal{C})$. Then, a the *reparameterized family* $\mathcal{R}_\Pi(Q)$ is:

$$r^*\Phi := \Phi \circ (\Delta r \times \mathrm{id}_X),$$

whose components are given by

$$(r^*\Phi)_s = \Phi_s \circ (r \times \mathrm{id}_{X(s)}).$$

Here, $\Delta r : \Delta Q \Rightarrow \Delta P$ is a natural transformation between the constant functors $\Delta Q, \Delta P : \mathcal{S} \to \mathcal{C}$. Δr has the same component $r : Q \to P$ at every object:

$$(\Delta r)_s := r$$

To quickly verify naturality, take any $u : s \to t$ in $\mathcal{S}$ and note that the following diagram commutes:

$$\begin{array}{ccc} P & \xrightarrow{\Delta Q(u)=\mathrm{id}_Q} & Q \\ {\scriptstyle (\Delta r)_s = r}\downarrow & & \downarrow{\scriptstyle (\Delta r)_t = r} \\ Q & \xrightarrow[\Delta P(u)=\mathrm{id}_P]{} & P \end{array}$$

Proposition 2.15 (Reparameterized families remain natural)**.** Given a realized family $\Phi \in \mathcal{R}_\Pi(P)$ and a reparameterization $r : P \to Q$, the reparameterized family $r^*\Phi$ is an element of $\mathcal{R}_\Pi(Q)$. In other words, $r^*\Phi$ is a natural transformation $Q \times X \Rightarrow Y$.

Proof. See Section A □

Thus, reparameterized families still respect the underlying prior Π. Moreover,

$$(\mathrm{id}_P)^*\Phi = \Phi \circ (\Delta \mathrm{id}_P \times \mathrm{id}_X) = \Phi,$$

and, for some $q : R \to Q$.

$$\begin{aligned} q^*(r^*\Phi) &= q^*(\Phi \circ (\Delta r \times \mathrm{id}_X)) \\ &= (\Phi \circ (\Delta r \times \mathrm{id}_X)) \circ (\Delta q \times \mathrm{id}_X) \\ &= \Phi \circ ((\Delta r \times \mathrm{id}_X) \circ (\Delta q \times \mathrm{id}_X) \quad \text{(Associativity of composition)} \\ &= (r \circ q)^*\Phi. \end{aligned}$$

Thus, we actually obtain a "realization functor" $\mathcal{R}_\Pi : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$, sending parameter spaces to their family of realizations, and reparameterizations to reparameterized families. We now use this construction to construct a category of realizations.

Definition 2.16 (Category of realizations)**.** Fix a prior Π. The *category of realizations* $\mathbf{Real}_{\mathcal{C}}(\Pi)$ consists of:

- **Objects:** Pairs (P, Φ), where $P \in \mathcal{C}$ is a parameter object, and $\Phi \in \mathcal{R}_\Pi(P)$ is a realized family.
- **Morphisms:** A morphism $r : (Q, \Psi) \to (P, \Phi)$ is an arrow $r : Q \to P$ in $\mathcal{C}$ such that:

$$\Psi = r^* \Phi.$$

- **Identity and composition:** Those of the underlying parameter maps in $\mathcal{C}$.

For a proof of associativity and unity, see Section A.

Intuitively, $\mathbf{Real}_{\mathcal{C}}(\Pi)$ is a category of parametric maps that respects Π. This mirrors one of our intuitions about how priors affect the space of admissible models: It generally does not restrict the space so much so that it uniquely picks out a single model. Instead, a prior induces a space of admissible models, and it does so through the soon-to-be-defined "realization functor". It is worth noting here the variance in morphisms in $\mathbf{Real}_{\mathcal{C}}(\Pi)$: a reparameterization $Q \to P$ makes the Q-parameterized family factor through the P-parameterized one.

We now get to a central connection between priors and parameterized architectures.

Theorem 2.17 (The realization functor)**.** There exists a functor

$$\mathrm{Real}_{\mathcal{C}} : \mathbf{Prior}_{\mathcal{C}}^{\mathrm{op}} \to \mathbf{Cat},$$

defined by:

(i) On objects (priors). $\Pi \mapsto \mathbf{Real}_{\mathcal{C}}(\Pi)$.

(ii) On morphisms (prior morphisms). For $m = (F, a, b) : \Pi_1 \to \Pi_2$, define a functor

$$\mathbf{Real}_{\mathcal{C}}(m) : \mathbf{Real}_{\mathcal{C}}(\Pi_2) \to \mathbf{Real}_{\mathcal{C}}(\Pi_1)$$

by:

(a) on objects: $(P, \Phi) \mapsto (P, T_m^P \Phi)$, where

$$T_m^P \Phi := b \circ (\Phi F) \circ (\mathrm{id}_{\Delta P} \times a), \qquad (T_m^P \Phi)_s := b_s \circ \Phi_{F(s)} \circ (\mathrm{id}_P \times a_s),$$

(b) on morphisms: $r \mapsto r$ (reparameterizations are sent to themselves).

Moreover, T_m^P is natural in reparameterizations:

$$\begin{array}{ccc} \mathcal{R}_{\Pi_2}(P) & \xrightarrow{T_m^P} & \mathcal{R}_{\Pi_1}(P) \\ \Big\downarrow{\scriptstyle r^*} & & \Big\downarrow{\scriptstyle r^*} \\ \mathcal{R}_{\Pi_2}(Q) & \xrightarrow[T_m^Q]{} & \mathcal{R}_{\Pi_1}(Q) \end{array}$$

Proof. Given a prior morphism $m = (F, a, b) : \Pi_1 \to \Pi_2$, define transport of realizations by

$$(T_m^P \Phi)_s = b_s \circ \Phi_{F(s)} \circ (\mathrm{id}_P \times a_s).$$

Naturality of a, b, and Φ ensures that the transported family respects Π_1. Moreover, transport commutes with reparameterization:

$$T_m^Q(r^*\Phi) = r^*(T_m^P \Phi).$$

Consequently, transport defines a functor between the corresponding realization categories. Compatibility with identities and composition of prior morphisms establishes the claimed contravariant functoriality. For a full proof, see Section A. □

Theorem 2.17 says that given a family Φ realizing Π_2, and a prior morphism $m = (F, a, b) : \Pi_1 \to \Pi_2$, the transported family Ψ is

$$\Psi(p, x) = b_s(\Phi_{F(s)}(p, a_s(x))).$$

We have constructed a direct bridge to the category of parameterized maps $\mathbf{Para}(\mathcal{C})$ [Gavranović, 2024], since at a particular realization $(P, \Phi) \in \mathbf{Real}_{\mathcal{C}}(\Pi)$, we can evaluate Φ at an object $s \in \mathcal{S}$ to get a map $\Phi_s : P \times X(s) \to Y(s)$. This is exactly the data of a parametric morphism $(P, \Phi_s) : X(s) \to Y(s)$ in $\mathbf{Para}(\mathcal{C})$. In other words, there is an evaluation:

$$\Pi \xrightarrow{\text{Real}} \mathbf{Real}_{\mathcal{C}}(\Pi) \xrightarrow{\text{ev}_s} \mathbf{Para}(\mathcal{C})$$

The prior does not canonically choose the Para morphism. Instead, it determines the category from which admissible Para morphisms may be selected. The maps $X(u) : X(s) \to X(t)$ and $Y(u) : Y(s) \to Y(t)$ technically become parameterized maps with terminal parameter object $1 \in \mathcal{C}$, as for any other non-parametric morphism in $\mathbf{Para}$.

Consider again $\mathcal{C} = \mathbf{Set}$. Then,

$$\mathcal{R}_\Pi(P) = \mathrm{Nat}(\Delta P \times X, Y) \cong \mathbf{Set}(P, \mathcal{H}_\Pi).$$

A realization is exactly a parameter-indexed collection of admissible models. Transport applies the existing admissible-model map T_m to every element of that collection. The categorical result organizes this uniformly across parameter spaces and maps between them. The theorem guarantees that this adapted implementation, $T_m^P \Phi \in R_{\Pi_1}(P)$, satisfies Π_1 for every parameter value. That is, we can reuse the original parameterization and its parameter sharing. This is what "natural in reparameterizations" mean; take a map $r : Q \to P$, then the transported family is

$$(r^*\Phi)_s(q, x) = \Phi_s(r(q), x).$$

Now, either reparameterizing and then transporting, or transporting first and then reparameterization both give the same result.

Remark 2.18. The assumption that Φ realizes a prior using P already means that the relevant structural equation holds for every parameter value $p \in P$. Consequently, it doesn't matter if

we have random parameters or trained parameters, each create models that stay within the structurally admissible family. As long as parameter values remain in P and the implementation remains the specified family, then no admissibility is broken. However, training could improve accuracy, destroy accuracy, or do nothing, still while structural admissibility survives. This does not mean the predictions are accurate. The chosen prior might be completely irrelevant, too general, or too restrictive. We have not made any claims about generalization capability or performance.

3 Applications

3.1 An equivalence between two geometric priors

In **?**, group equivariance is studied logically using first-order logic

Fix a group G and take our base category to be $\mathcal{C} = \mathbf{Set}$. Consider the prior

$$\Pi_{\mathbb{L}G} := (\mathbb{L}_G^{\text{act}}, X_1, Y_1),$$

with the prior specification consisting of the Lawvere theory of actions of G, denoted $\mathbb{L}_G^{\text{act}}$. This is the algebraic theory generated freely by the following operations and axioms:

$$\Sigma = \{s_g : 1 \to 1 \quad \forall g \in G\}, \quad E = \begin{Bmatrix} s_g \circ s_h = s_{gh} & \forall g, h \in G \\ s_e = \text{id}_1 & e \in G \end{Bmatrix}.$$

Then,

$$\mathbb{L}_G^{\text{act}} = \text{Free}_{\times}(\Sigma)/\sim_E .$$

The input and output adapters $X_1, Y_1 : \mathbb{L}_G^{\text{act}} \to \mathbf{Set}$ are then technically algebras of $\mathbb{L}_G^{\text{act}}$, defined as:

$$\begin{aligned} X_1(1) &:= \mathbb{R}^m, & Y_1(1) &:= \mathbb{R}^n, \\ X_1(s_g) &:= \rho_{\mathbb{R}}^n(g), & Y_2(s_g) &:= \rho_{\mathbb{R}}^m(g). \end{aligned}$$

We can prove functoriality of X by using the axioms of the algebraic theory and the axioms of a group action $\rho : \mathbb{R}^{\ltimes} \to \mathbb{R}^n$:

$$\begin{aligned} X(s_g \circ s_h) = X(s_{gh}) = \rho_{\mathbb{R}^n}(gh) = \rho_{\mathbb{R}^n}(g) \circ \rho_{\mathbb{R}^n}(h) = X(s_g) \circ X(s_h), \\ X(\text{id}_1) = X(s_e) = \rho_{\mathbb{R}^n}(e) = \text{id}_{\mathbb{R}}. \end{aligned}$$

The prior $\Pi_{\mathbb{L}G}$ is not very interesting to consider in isolation, but we note that $\mathcal{H}_{\Pi_{\mathbb{L}G}}$ describes the set of equivariant models. In this case, since Lawvere theories have finite products, $n \cong 1^n$, and since algebras X and Y are product-preserving,

$$X(n) \cong X(1^n) = X(1)^n.$$

Likewise for Y. Thus, $\mathcal{H}_{\Pi_{\mathbb{L}G}}$ is the set of predictors $\{h_1 : X(1) \to Y(1)\}$ such that:

$$\begin{array}{ccc} X(1) & \xrightarrow{X(s_g)} & X(1) \\ {\scriptstyle h_1}\downarrow & & \downarrow{\scriptstyle h_1} \\ Y(1) & \xrightarrow[Y(s_g)]{} & Y(1) \end{array} \quad = \quad \begin{array}{ccc} \mathbb{R}^m & \xrightarrow{\rho_{\mathbb{R}^m}(g)} & \mathbb{R}^m \\ {\scriptstyle h_1}\downarrow & & \downarrow{\scriptstyle h_1} \\ \mathbb{R}^n & \xrightarrow{\rho_{\mathbb{R}^m}(g)} & \mathbb{R}^n \end{array}$$

Unpacking this element-wise yields group equivariance:

$$h_1(g \cdot x) = g \cdot h_1(x).$$

So $\mathcal{H}_{\Pi_{\mathbb{L}G}}$ really describes the "set of models that respect the prior".

Using Lawvere theories is certainly not the only way to mathematically represent groups. In Maruyama [2025], the same notion of geometric priors is expressed using the one-object category BG. Fix a group G, then BG consists of one object G with morphisms $s_g : G \to G$ for each $g \in G$, with the axioms of a group quotiented out. A functor $X : BG \to \mathcal{C}$ then specifies an object $X \in \mathcal{C}$ together with an endomorphism $g_* : X \to X$ for all $g \in G$. Functoriality of X gives the same two conditions we've seen:

- $h_* g_* = (hg)_*$ for all $g, h \in \mathcal{C}$.
- $e_* = \mathrm{id}_X$, where $e \in G$ is the identity element.

The functor X defines an action of G on X. (!)

Fix a group G and let BG be its one-object category representation. Then, we can describe a geometric prior; consider two spaces $A, B \in \mathcal{C}$ and let $\rho_A : G \to \mathrm{Aut}(A)$ and $\rho_B : G \to \mathrm{Aut}(B)$ describe group actions of G on A and B, respectively. These actions induce two functors

$$\begin{aligned} A_G : BG \to \mathcal{C}, \quad A_G(G) = A, \qquad A_G(s_g) = \rho_A(g), \\ B_G : BG \to \mathcal{C}, \quad B_G(G) = B, \qquad B_G(s_g) = \rho_B(g), \end{aligned}$$

Thus, we get a prior $\Pi_{BG} = (BG, A_G, B_G)$. Then, an element of $\mathcal{H}_{\Pi_{BG}}$ has one component $h : A \to B$, whose naturality equations are precisely equivariance. There exists a functor $J : BG \to \mathbb{L}_G^{\mathrm{act}}$ defined by $* \mapsto 1$ and $g \mapsto s_g$. Because $Xj = A_G$ and $Bj = B_G$, we get a prior morphism:

$$m_j = (j, \mathrm{id}, \mathrm{id}) : \Pi_{BG} \to \Pi_{\mathbb{L}_G^{\mathrm{act}}}.$$

Its induced map on admissible models is the restriction:

$$T_{m_j} : \mathrm{Nat}(X, Y) \longrightarrow \mathrm{Nat}(A_G, B_G), \quad h \mapsto h_1.$$

However, this restriction is bijective. !? Therefore,

$$\mathrm{Nat}(X, Y) \cong \mathrm{Nat}(A_G, B_G),$$

yet the specification categories themselves are not isomorphic! A category isomorphic to a one-object category has only one isomorphism class of objects. But a Lawvere theory has objects 0 and 1 that are not isomorphic. But the priors Π_{BG} and $\Pi_{\mathbb{L}_G^{\mathrm{act}}}$ are equivalent by presentation.

3.2 Invariance and equivariance

A core part of the definition of the prior is that the input and output representation of the prior is part of the prior itself. This means morphisms of priors are not just morphisms between prior specifications, but between prior representations as well. Let $\Pi_{\mathbb{L}G} := (\mathbb{L}_G^{\text{act}}, X, Y)$ be as in Section 3.1. Next, consider a second prior $\Pi'_{\mathbb{L}G} = (\mathbb{L}_G^{\text{act}}, X, T)$ where $T : \mathbb{L}_G^{\text{act}} \to \mathbf{Set}$ is a "trivial" output functor defined as:

$$\begin{aligned} T(1) &:= \mathbb{R}^n, \\ T(s_g) &:= \mathrm{id}_{\mathbb{R}^n}. \end{aligned}$$

Functoriality of T holds since:

$$\begin{aligned} T(s_g \circ s_h) = X(s_{gh}) = \mathrm{id}_{\mathbb{R}^n} = \mathrm{id}_{\mathbb{R}^n} \circ \mathrm{id}_{\mathbb{R}^n} = T(s_g) \circ T(s_h), \\ T(\mathrm{id}_1) = T(s_e) = \mathrm{id}_{\mathbb{R}^n}. \end{aligned}$$

$\mathcal{H}_{\Pi'_{\mathbb{L}G}}$ consists of predictors that satisfy invariance, which, element-wise, after unpacking the naturality square becomes:

$$h_1(g \cdot x) = \mathrm{id}_{\mathbb{R}^n} \circ h(x) = h(x).$$

Now, there exists a morphism from the invariance prior to the equivariance prior, namely

$$\phi := (\mathrm{id}_{\mathcal{S}}, \mathrm{id}_X, q) : \Pi'_{\mathbb{L}G} \to \Pi_{\mathbb{L}G},$$

where $q : T \Rrightarrow Y$ is determined by $q_1 : \mathbb{R}^n \to \mathbb{R}^n$, which we define as:

$$q_1(\rho(g)x) = q_1(x).$$

This satisfies naturality:

$$(q_1 \circ \rho_{\mathbb{R}^n}(g))(x) = q_1(\rho_{\mathbb{R}^n}(g)(x)) = q_1(x) = \mathrm{id}_{\mathbb{R}^n} \circ q_1(x).$$

By Theorem 2.7, ϕ induces a contravariant function

$$T_\phi : \mathrm{Nat}(X, T) \longrightarrow \mathrm{Nat}(X, Y), \quad T_\phi(h) = q \circ (h\mathrm{id}_{\mathcal{S}}) \circ \mathrm{id}_X = q \circ h.$$

This means prior morphisms can express output postprocessing, and here an equivariant predictor is turned into an invariant one by an invariant readout q. To take it down one more level of abstraction, let $G = \{e, \sigma\}$ act on $\mathbb{R}^2$ by swapping coordinates. That is, $\sigma(x_1, x_2) := (x_2, x_1)$ and $\rho(g) = \sigma$. We can for example choose $q(y_1, y_2) = y_1 + y_2$, which is invariant because

$$q(\sigma(y_1, y_2)) = q(y_2, y_1) = y_2 + y_1 = q(y_1, y_2).$$

Now, consider a neural network $h : \mathbb{R}^2 \to \mathbb{R}$ that respects this specific instance of $\Pi_{\mathbb{L}G}$. Then, for $f := q \circ h : \mathbb{R}^2 \to \mathbb{R}$,

$$\begin{aligned} f(\sigma x) &= q(h(\sigma x)) \\ &= q(\sigma h(x)) \\ &= q(h(x)) \\ &= f(x). \end{aligned}$$

So f is invariant.

This highlights the semantic content of the prior morphism; a morphism from the equivariance prior $\Pi_{\mathbb{L}G}$ to the invariance prior $\Pi'_{\mathbb{L}G}$ induces a contravariant function which maps any equivariant model to its invariant model. The distinction between prior morphism and the induced admissible model morphism is important to note here.

The prior morphism ϕ in **??** should not be misinterpreted as saying that "all equivariant models are invariant", which isn't true. Instead, it should be interpreted as saying that every equivariant predictor h can be transformed into an invariant predictor f, provided we have chosen an invariant output map q, by precomposing with q.

3.3 Groupoids and geometric priors

Building on the work in Maruyama [2025, Corollary 2], we ask whether there exists a useful prior morphism between Π_{BG} and a certain *groupoid prior* $\Pi_{\mathcal{G}}$. Let $\mathcal{G}$ be a groupoid. Then the groupoid prior studied is $\Pi_{\mathcal{G}} = (\mathcal{G}^{op}, X, Y)$ over **Top**. To construct the prior morphism we are interested in, choose an object $s \in \mathcal{G}$, and let $H = \mathrm{Hom}_{\mathcal{G}}(s)$ be its isotropy group. Then, there is an inclusion functor $i : BH \hookrightarrow \mathcal{G}$ with $* \mapsto s$ and each group element to its corresponding loop. Define $\Pi_{BH} = (BH, Xi, Yi)$, then we immediately obtain a prior morphism $(i, \mathrm{id}, \mathrm{id}) : \Pi_{BH} \to \Pi_{\mathcal{G}}$. The action on admissible models maps $h \mapsto h_s$. For a connected groupoid, we get an equivalence of priors. There would exist arrows

$$t_a : s \to a, \quad t_s = \mathrm{id}_s.$$

Now, we can define a prior morphism in the other direction:

$$r : \mathcal{G} \to BH, \quad r(u) = t_a^{-1} u t_b, \quad u : b \to a.$$

Then,

$$n = (r, \alpha, \beta) : \Pi_{\mathcal{G}} \to \Pi_{BH}$$

exists with

$$\alpha_a = Xi(t_a^{-1}), \quad \beta_a = Yi(t_a).$$

Then, its induced map on admissible models is

$$h_a = Y(t_a) \circ f \circ X(t_a)^{-1}.$$

Thus we have prior morphisms in both directions whose induced maps on admissible models are inverse, so

$$\Pi_{\mathcal{G}} = \Pi_{BH}.$$

Here we see that models respecting the groupoid prior gives an ordinary H-equivariant model by evaluation at s.

3.4 Causal priors in Transformers

Imagine a streaming speech recognizer. Its output at time t must be determined by the audio received up to t. Let A be the space of input samples and B the space of outputs emitted at each step. Define a prior $\Pi_{\text{stream}} = (\mathcal{T}, X, Y)$, where $\mathcal{T}$ is the poset category on $(\mathbb{N}, \leq)$, $X(n) = A^n$ with arrows $n \to m$ interpreted as prefix truncations, meaning that we map it to some truncation $A^n \to A^m$ by truncating $(a_1, a_2, \ldots, a_m, \ldots, a_n)$ to $(a_1, a_2, \ldots, a_m)$. An admissible model is a family of component $h_n : A^n \to B^n$ satisfying

$$h_n(x_1, \ldots, x_n) \mid_{1:m} = h_m(x_1, \ldots, x_m).$$

Intuitively, extending the input cannot revise an already emitted output. This is a non-anticipation constraint for the resulting stream map.

This sort of prior modeled using poset specifications is common across non-anticipation tasks, such as in attention mechanisms, and WaveNets.

Let A be the input-token or input-feature space, and B the per-position output space. For language modeling, B could be the space of next-token probability distributions. Define a category $\mathcal{S}_{\text{prefix}}$ with objects corresponding to sequences of lengths $n \in \mathbb{N}$, and with a morphism $n \to m$ existing whenever $m \leq n$, representing truncation to the first m positions. Define X and Y as in **??**. Then, let $\Pi_{\text{causal}} = (\mathcal{S}_{\text{prefix}}, X, Y)$. An admissible model consists of functions $f_n : A^n \to B^n$ satisfying

$$\pi^B_{n,m} \circ f_n = f_m \circ \pi^A_{n,m}.$$

Intuitively, processing the whole sequence and keeping its first m outputs, or first truncating the input and then processing it should yield the same result. This is one prior that the attention mechanism in the original transformer respects, where future text cannot change earlier predictions. Indeed, taking $m = i$ gives

$$[f_n(x_1, \ldots, x_n]_i = [f_i(x_1, \ldots, x_i)]_i.$$

The right-hand side has no access to $x_{i+1}, \ldots, x_n$[4]. Consider an attention layer with the same parameters at every supported sequence length. At position i,

$$o_i = \frac{\sum_{j=1}^{i} \exp(q_i^\top k_j / \sqrt{d_k}) v_j}{\sum_{j=1}^{i} \exp(q_i^\top k_j / \sqrt{d_k})}.$$

Appending tokens after i changes neither the terms nor the denominator. Therefore the output at i stays unchanged. Shared positionwise feed-forward layers, ordinary per-token normalization, and residual connections preserve this property. Consequently, their composition does too. This gives a proof for the whole transformer by checking its components.

3.5 Logical priors in boolean networks

$$\Sigma = \{\wedge : 2 \to 1,\ \vee : 2 \to 1,\ \neg : 1 \to 1\}, \tag{1}$$

$$\Phi = \{\neg \circ \wedge = \vee \circ (\neg \times \neg)\}. \tag{2}$$

[4]Note that this prior expresses both non-anticipation and consistency across sequence lengths. Separate functions f_n could each avoid future tokens while still change their behavior depending on the total length n. Naturality rules that out too.

4 Experimentation

4.1 Modular Arithmetic

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A Proofs

Proof of associativity, unity, and local-smallness in Definition 2.6. We need to show that $\mathbf{Prior}(\mathcal{C})$ satisfies associativity and unity. First, the composition $n \circ m$ is well-typed since:

$$X_1 \overset{a}{\Rightarrow} X_2F \overset{cF}{\Rightarrow} X_3GF,$$
$$Y_3GF \overset{dF}{\Rightarrow} Y_2F \overset{b}{\Rightarrow} Y_1,$$

and whiskering and composition preserve naturality. To show associativity, let $r := (H, e, k) : \Pi_3 \to \Pi_4$. By definition,

$$r \circ (n \circ m) = (H(GF), (e(GF)) \circ ((cF) \circ a), (b \circ (dF)) \circ (k(GF))),$$
$$(r \circ n) \circ m = ((HG)F, (((eG) \circ c)F) \circ a, b \circ ((d \circ (kG))F)).$$

Whiskering distributes over composition, and repeated whiskering agrees with whiskering by a composite. Thus both triples are exactly

$$(HGF, (eGF) \circ (cF) \circ a, b \circ (dF) \circ (kGF)).$$

Functor composition and natural transformation composition is associative. For units, direct substitution yields:

$$1_{\Pi_2} \circ m = (F, (\mathrm{id}_{X_2} F) \circ a, b \circ (\mathrm{id}_{Y_2} F)) = (F, a, b),$$
$$m \circ 1_{\Pi_1} = (F, (a\mathrm{id}_{\mathcal{S}_1}) \circ \mathrm{id}_{X_1}, \mathrm{id}_{Y_1} \circ (b\mathrm{id}_{\mathcal{S}_1})) = (F, a, b).$$

□

Proof of Proposition 2.9. If $F^{-1} = G$ and a, b are invertible, an inverse is given by $(G, (aG)^{-1}, (bG)^{-1})$. Conversely, suppose (G, c, d) is an inverse. Equality of underlying specification functors gives $GF = \mathrm{id}$ and $FG = \mathrm{id}$. The input comparison give $(cF)a = \mathrm{id}$ and $(aG)c = \mathrm{id}$. Whiskering the second by F gives $a(cF) = \mathrm{id}$, so $a^{-1} = cF$. The output equations give $b(dF) = \mathrm{id}$ and $d(bG) = \mathrm{id}$. Whiskering the latter by F gives $(dF)b = \mathrm{id}$. Thus $b^{-1} = dF$. □

Proof of Proposition 2.15. $\Delta r \times \mathrm{id}_X : \Delta Q \times X \Rightarrow \Delta P \times X$ has components

$$(\Delta r \times \mathrm{id}_X)_s = r \times \mathrm{id}_{X(s)} : Q \times X(s) \to P \times X(s).$$

It sends $(q, x) \mapsto (r(q), x)$, substituting the parameter and leaving the input itself unchanged. To check naturality, let $u : s \to t$ in $\mathcal{S}$. Then:

$$(\mathrm{id}_P \times X(u)) \circ (r \times \mathrm{id}_{X(s)}) = (\mathrm{id}_P \circ r) \times (X(u) \circ r) = r \times X(u),$$
$$(r \times \mathrm{id}_{X(t)}) \circ (\mathrm{id}_Q \times X(u)) = (r \circ \mathrm{id}_Q) \times (\mathrm{id}_{X(t)} \circ X(u)) = r \times X(u),$$

by the (cartesian) product-composition rule. Composing $\Delta r \times \mathrm{id}_X$ with Φ, which is natural by assumption, yields a vertical composite,

$$(\Phi \circ (\Delta r \times \mathrm{id}_X))_s = \Phi_s \circ (\Delta r \times \mathrm{id}_X)_s,$$

which is a natural:

$$\begin{aligned} Y(u) \circ \Phi_s \circ (\Delta r \times \mathrm{id}_X)_s &= \Phi_t \circ (\mathrm{id}_P \times X(u)) \circ (\Delta r \times \mathrm{id}_X)_s && \text{(naturality of } \Phi) \\ &= \Phi_t \circ (\Delta r \times \mathrm{id}_X)_t \circ (\mathrm{id}_Q \times X(u)). && \text{(naturality of } \Delta r \times \mathrm{id}_X) \end{aligned}$$

□

Proof of associativity and unity in Definition 2.16. Suppose

$$(R, \Omega) \xrightarrow{q} (Q, \Psi) \xrightarrow{r} (P, \Phi).$$

Then:

$$\Omega = q^* \Psi = q^* (r^* \Phi) = (r \circ q)^* \Phi$$

So $r \circ q : (R, \Omega) \to (P, \Phi)$ is indeed a valid morphism. Unity follows from

$$(\mathrm{id}_P)^* \Phi = \Phi.$$

Associativity and the unit laws follow from those in $\mathcal{C}$. □

Proof of Theorem 2.17. First, the composite $T_m^P\Phi$ has the required type. It is the composite

$$\Delta P \times X_1 \xrightarrow{\mathrm{id}_{\Delta P}\times a} \Delta p \times X_2 F \xrightarrow{\Phi F} Y_2 F \xrightarrow{b} Y_1.$$

To verify its naturality explicitly, fix $u : s \to t$ in $\mathcal{S}_1$. The naturality assumptions give

$$X_2(Fu) \circ a_s = a_t \circ X_1(u), \qquad Y_1(u) \circ b_s = b_t \circ Y_2(Fu),$$

and

$$Y_2(Fu) \circ \Phi_{F(s)} = \Phi_{F(s)} \circ (\mathrm{id}_P \times X_2(Fu)).$$

Consequently,

$$\begin{aligned} Y_1(u) \circ (T_m^P\Phi)_s &= Y_1(u) \circ b_s \circ \Phi_{F(s)} \circ (\mathrm{id}_P \times a_s) \\ &= b_t \circ Y_2(Fu) \circ \Phi_{F(s)} \circ (\mathrm{id}_P \times a_s) \\ &= b_t \circ \Phi_{F(t)} \circ (\mathrm{id}_P \times X_2(Fu)) \circ (\mathrm{id}_P \times a_s) \\ &= b_t \circ \Phi_{F(t)} \circ (\mathrm{id}_P \times a_t) \circ (\mathrm{id}_P \times X_1(u)) \\ &= (T_m^P\Phi)_t \circ (\mathrm{id}_P \times X_1(u)). \end{aligned}$$

Thus $T_m^P\Phi \in R_{\Pi_1}(P)$. Next we prove compatibility with reparameterization. For $r : Q \to P$, functoriality of products gives

$$(r \times \mathrm{id}_{X_2(Fs)}) \circ (\mathrm{id}_Q \times a_s) = (\mathrm{id}_P \times a_s) \circ (r \times \mathrm{id}_{X_1(s)}).$$

Both sides are simply the product morphism $r \times a_s$. Therefore,

$$\begin{aligned} \left(T_m^Q(r^*\Phi)\right)_s &= b_s \circ \Phi_{F(s)} \circ (r \times \mathrm{id}_{X_2(Fs)}) \circ (\mathrm{id}_Q \times a_s) \\ &= b_s \circ \Phi_{F(s)} \circ (\mathrm{id}_P \times a_s) \circ (r \times \mathrm{id}_{X_1(s)}) \\ &= \left(r^*(T_m^P\Phi)\right)_s. \end{aligned}$$

This proves the naturality square $r^*(T_m^P\Phi) = T_m^Q(r^*\Phi)$. In categorical language, T_m is a natural transformation

$$T_m : R_{\Pi_2} \Rightarrow R_{\Pi_1}$$

between functors $\mathcal{C}^{op} \to \mathbf{Set}$. This also proves that transport is well-defined on realization morphisms. Indeed, if

$$r : (Q, \Psi) \to (P, \Phi), \qquad \Psi = r^*\Phi,$$

then

$$T_m^Q\Psi = T_m^Q(r^*\Phi) = r^*(T_m^P\Phi).$$

Hence the same r defines a morphism

$$(Q, T_m^Q\Psi) \longrightarrow (P, T_m^P\Phi).$$

Since transport leaves underlying parameter maps unchanged, it preserves their identities and composition. Thus each $\mathrm{Real}_{\mathcal{C}}(m)$ is a functor. It remains to verify functoriality in the prior. For the identity prior morphism,

$$T_{1_\Pi}^P\Phi = \Phi.$$

It also leaves every realization morphism unchanged, so

$$\mathrm{Real}_{\mathcal{C}}(1_{\Pi}) = \mathrm{id}_{\mathrm{Real}_{\mathcal{C}}(\Pi)}.$$

For composable prior morphisms

$$\Pi_1 \xrightarrow{m=(F,a,b)} \Pi_2 \xrightarrow{n=(G,c,d)} \Pi_3$$

and $\Psi \in R_{\Pi_3}(P)$, we have

$$\begin{aligned}
\left(T_m^P(T_n^P\Psi)\right)_s &= b_s \circ d_{F(s)} \circ \Psi_{GF(s)} \circ (\mathrm{id}_P \times c_{F(s)}) \circ (\mathrm{id}_P \times a_s) \\
&= b_s \circ d_{F(s)} \circ \Psi_{GF(s)} \circ \big(\mathrm{id}_P \times (c_{F(s)} \circ a_s)\big) \\
&= \left(T_{n\circ m}^P\Psi\right)_s.
\end{aligned}$$

The two constructions again leave every parameter map unchanged. Therefore,

$$\mathrm{Real}_{\mathcal{C}}(n \circ m) = \mathrm{Real}_{\mathcal{C}}(m) \circ \mathrm{Real}_{\mathcal{C}}(n).$$

The reversal of composition is precisely contravariant functoriality. These are equalities of functors, not merely isomorphisms. □